

One drone, four opening sizes

The imported MK4 assembly has a **401.3 mm full propeller-sweep width**. A **1000 mm opening** is the suggested next trial, with 299 mm per side on a centered straight approach. The current opening gives 540 mm per side. The 800 mm option is tighter at 199 mm per side.

Opening (mm)	Outer (mm)	Side gap (mm)	Vertical gap (mm)
1480.8	2700.0	540	656
1200.0	2419.2	399	515
1000.0	2219.2	299	415
800.0	2019.2	199	315

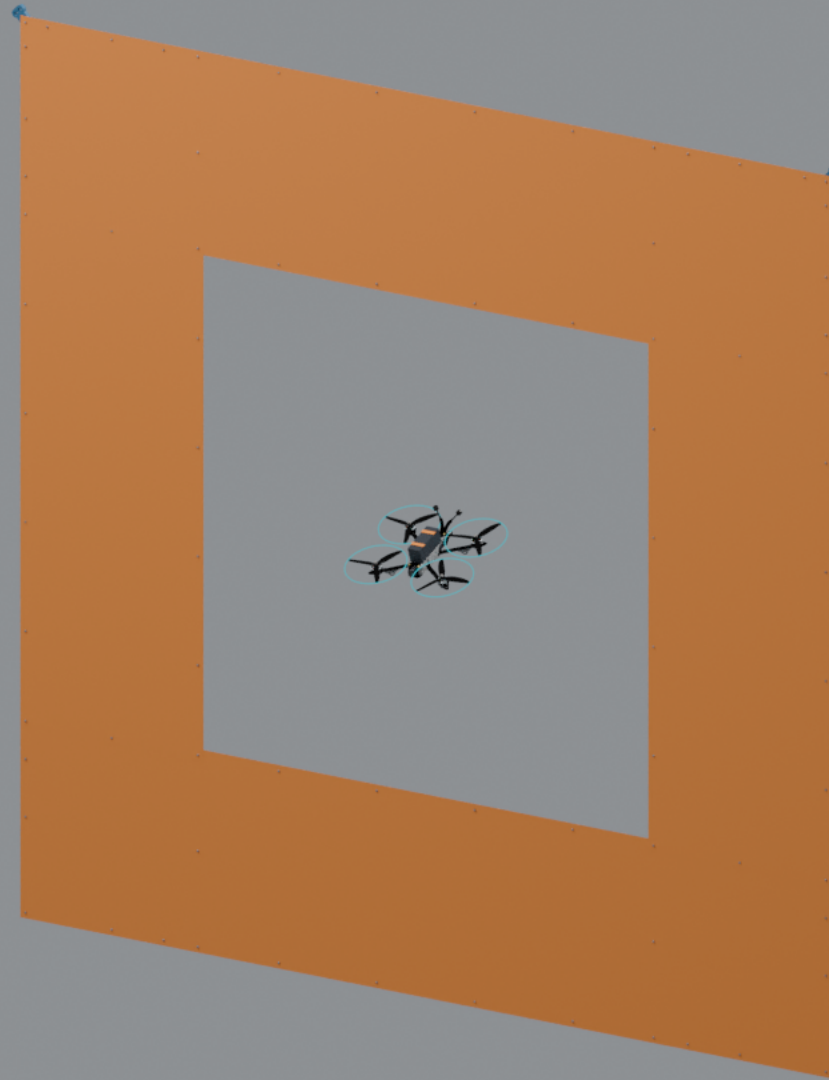
All gaps are **per edge**, at 20-degree forward pitch, zero yaw and roll. The projected envelope is centered. Battery dimensions (110 x 40 x 45 mm) are assumed; confirm against your build. These are geometric gaps, not guaranteed flight margins.

Keep the existing printed fittings and 24-inch paper bands. Replace eight long PVC members: 1560.8 mm current, 1280 mm for 1200 opening, 1080 mm for 1000, or 880 mm for 800. Sixteen short 389.6 mm members remain. Dry-fit before batch cutting. Full methods and paper cuts: accompanying README.md.

Drone reference: Dendy / Printables 1515387, CC BY-NC 4.0; the uploader disclaims original authorship of the included MK4 frame. Original gate material retains the separate project license. Cyan rings are prop-sweep envelopes, not physical guards.

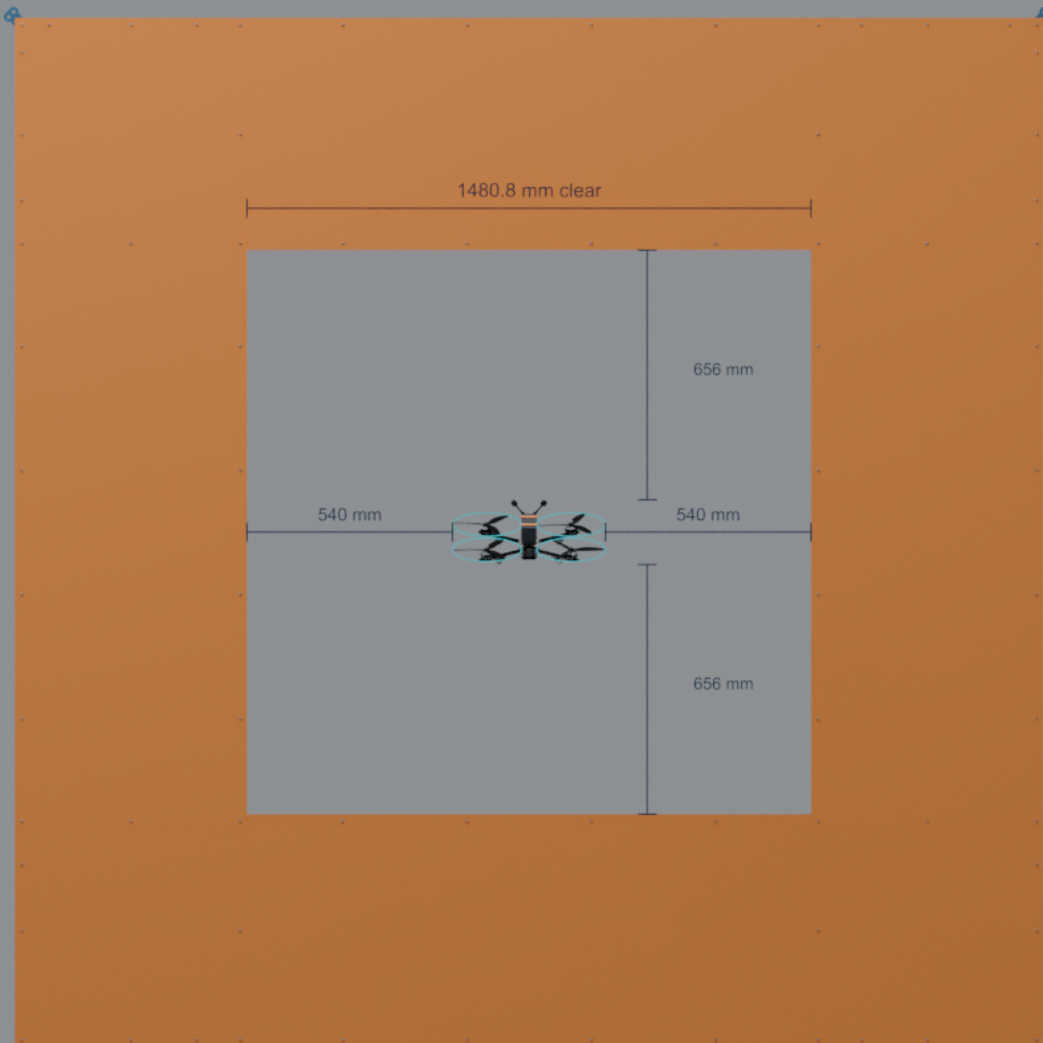
Your 7-inch quad / current gate

Imported MK4 assembly, centered at the crossing plane. 20-degree forward pitch; full swept propeller envelope.



Current opening / centered clearances

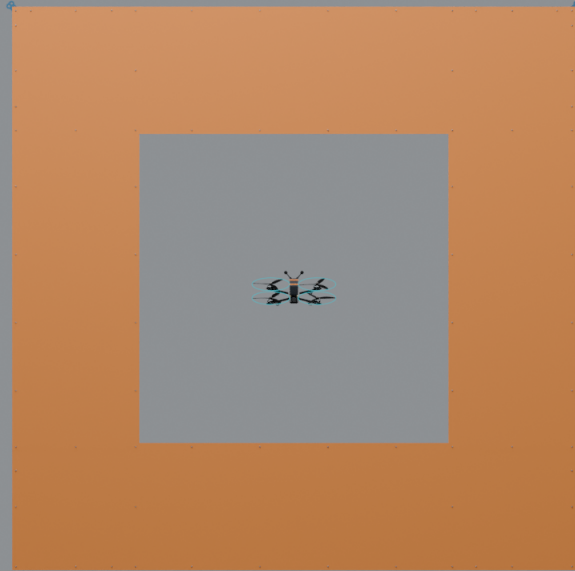
Swept width 401.3 mm; projected height 169.7 mm at 20-degree pitch. The reference envelope is centered in the opening.



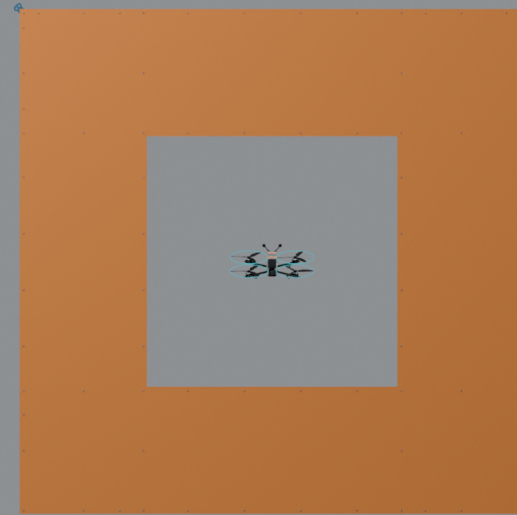
Static geometric clearance to the paper edges, not an allowance for tracking error, wind or paper movement. Cyan outlines = swept props.

Four openings / same drone / same scale

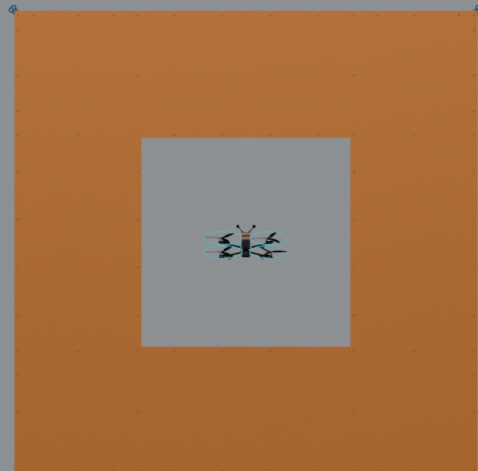
24-inch paper bands and existing fittings retained. Compare the clear opening, not the overall gate size.



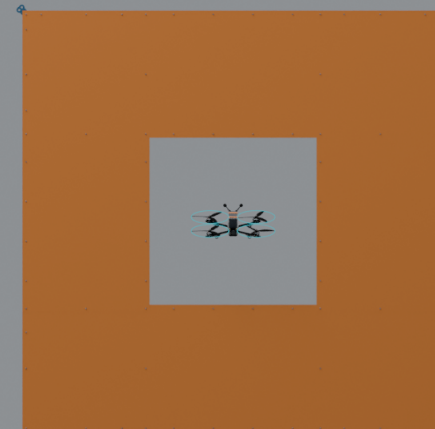
1481 mm opening | 540 mm each side



1200 mm opening | 399 mm each side



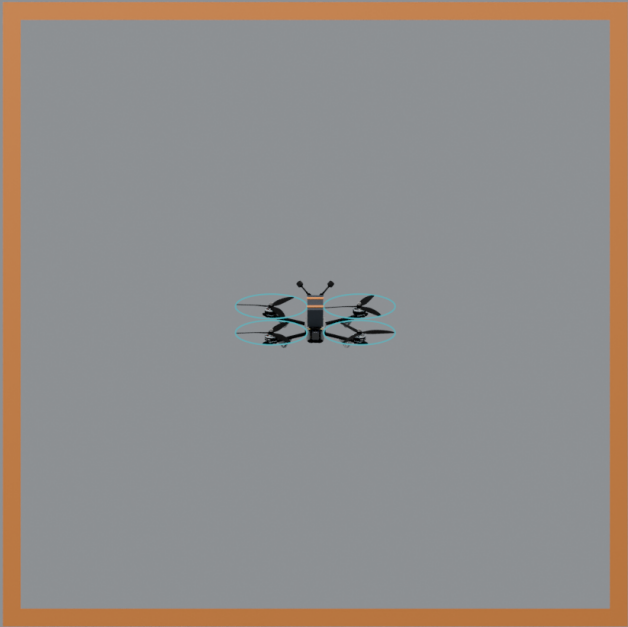
1000 mm opening | 299 mm each side



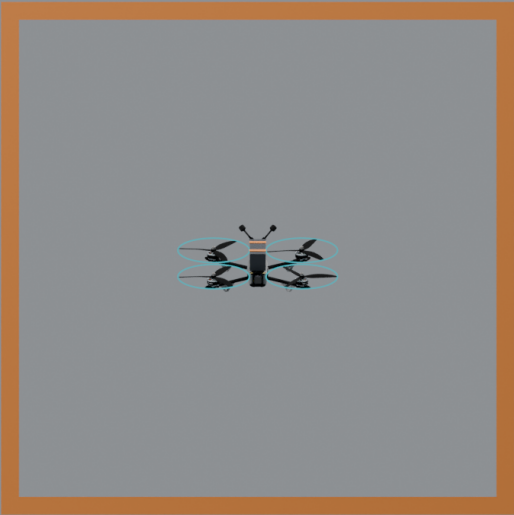
800 mm opening | 199 mm each side

Opening close-ups / equal scale

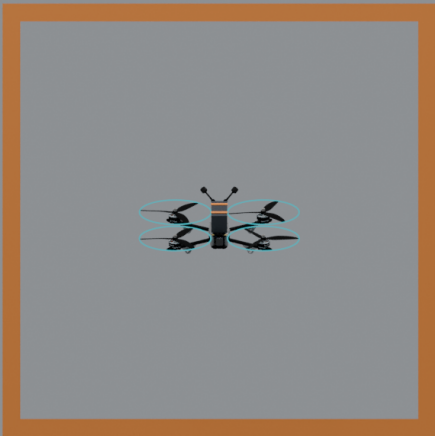
Only a narrow strip of each paper edge is shown. Drones remain the same size; pose is 20-degree forward pitch.



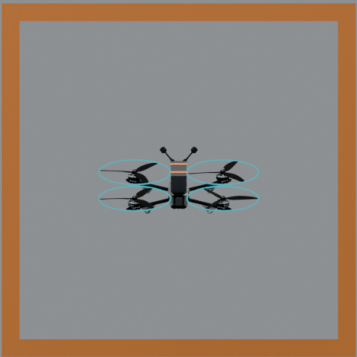
1481 mm / side gap 540 mm



1200 mm / side gap 399 mm



1000 mm / side gap 299 mm



800 mm / side gap 199 mm

Imported MK4 / swept propeller envelope

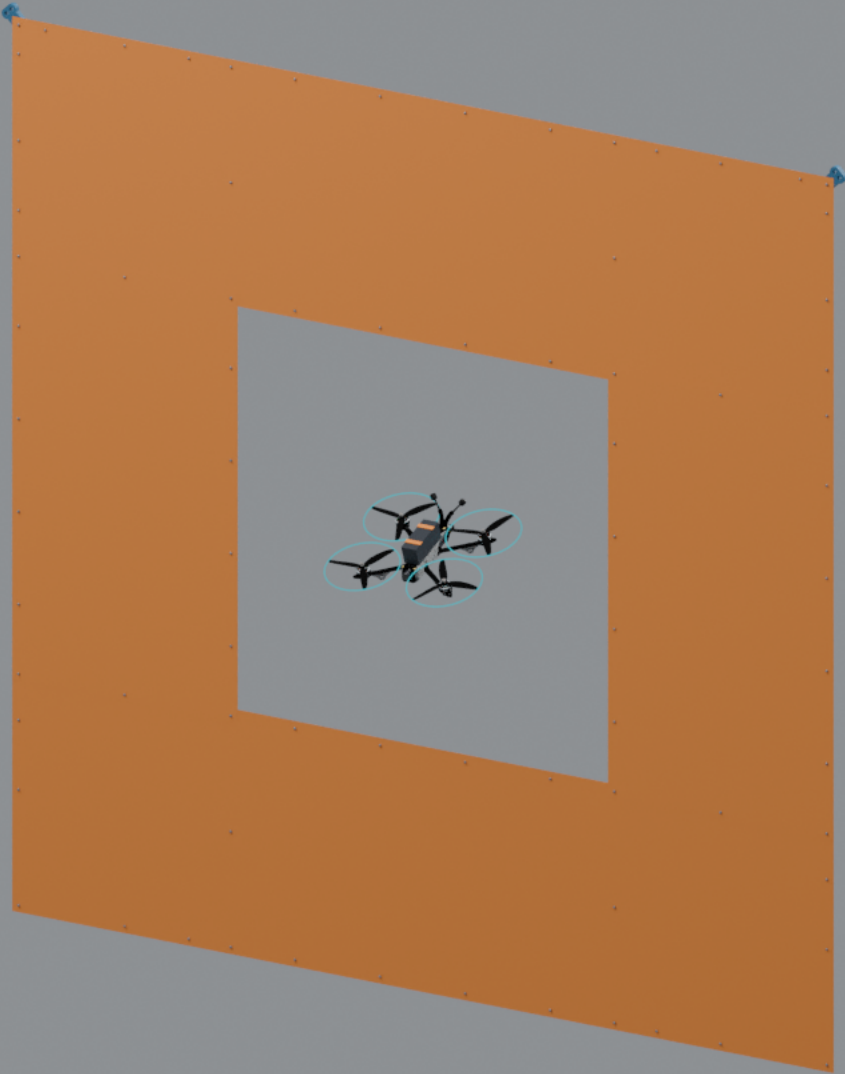
Motor diagonal 294.3 mm. CAD prop sweep diameter 179.9 mm; nominal seven inches = 177.8 mm.



Actual supplied assembly, reoriented upright. Cyan = prop sweep. Dark top battery with orange straps is an assumed addition.

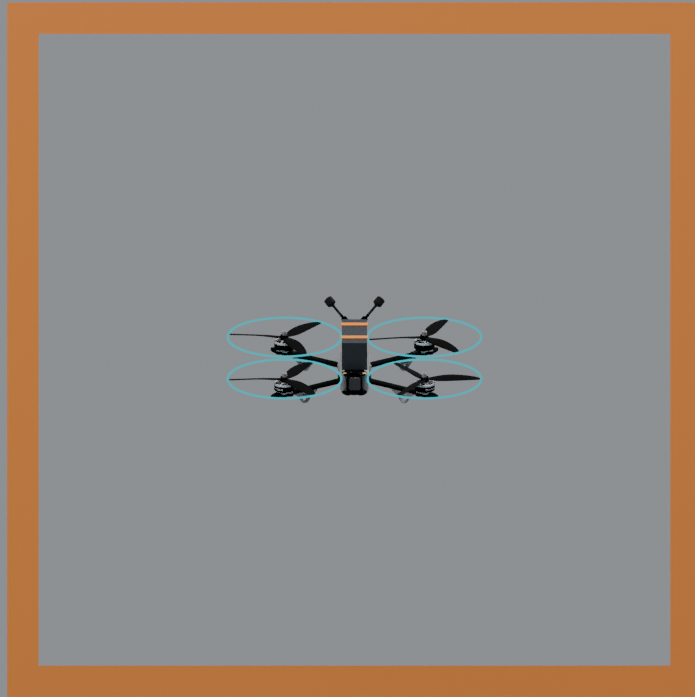
1000 mm opening / suggested next trial

Outer size 2219.2 mm with full-width paper. Centered side clearance: 299 mm per side at this pose.



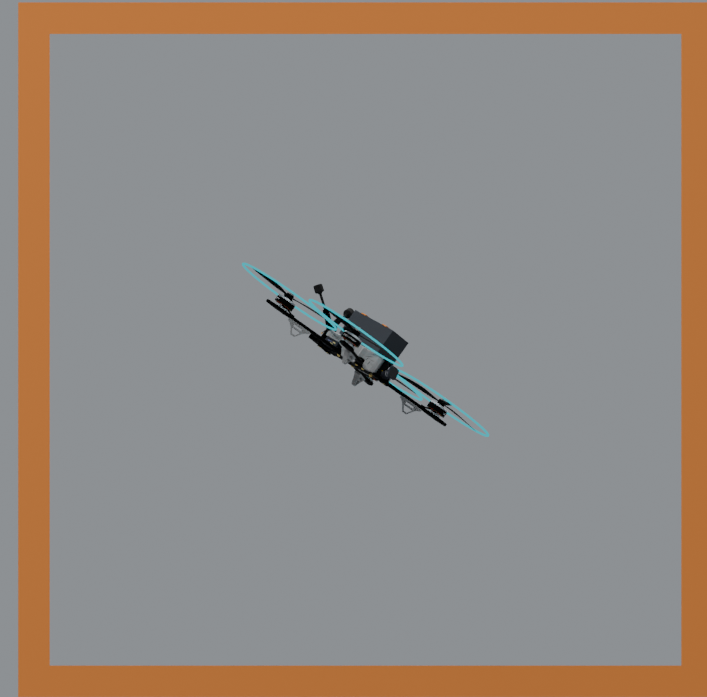
Attitude matters / both openings are 1 metre

The drone is re-centered by its projected swept envelope in each view. A turn or bank changes both clearances.



Yaw 0 / roll 0 / pitch 20 degrees

401 x 170 mm projected envelope



Yaw 45 / roll 30 / pitch 20 degrees

392 x 279 mm projected envelope

No flight-control error margin is included. Measure your actual props, battery and antennas before choosing an opening.